Verification of Solutions

Q1. Verification. For each of the following differential equations, verify that the given function is a solution (i.e. substitute the function into both sides of the differential equation). The given functions are the *general solutions* of the differential equations, so C is a constant of integration.

(a) $\dfrac{dy}{dx} = 1,\ y = x + C$

(b) $\dfrac{dy}{dx} = y,\ y = Ce^x$

(c) $\dfrac{dy}{dx} = 2y - 4x,\ y = Ce^{2x} + 2x + 1$

(d) $\dfrac{dy}{dx} = -\dfrac{2y}{x},\ y = \dfrac{C}{x^2}$

Q2. Constant Solutions. Find the *constant solutions* for the following separable differentiable equations.

(a) $\dfrac{dy}{dx} = e^x y.$

(b) $\dfrac{dy}{dx} = xe^y.$

(c) $\dfrac{dy}{dx} = \sin(x)(y^2 + 1).$

(d) $\dfrac{dy}{dx} = (x^2 - 1)\sin(2y).$

Q3. Initial Value Problems. Solve the following initial value problems, by starting with the general solutions verified in the Question 1 (so you do not need to solve them from scratch).

(a) $\dfrac{dy}{dx} = 1$, with $y(0) = 3.$

(b) $\dfrac{dy}{dx} = y$, with $y(0) = 2.$

(c) $\dfrac{dy}{dx} = 2y - 4x$, with $y(-1) = 0.$

(d) $\dfrac{dy}{dx} = -\dfrac{2y}{x}$, with $y(2) = 64$

Q4. Second Order. Verify that the given function is a solution of the given second order differential equation.

(a) $\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx} - 6y = 0$, $y = e^{-2x} + e^{3x}$

(b) $\dfrac{d^2y}{dt^2} = -n^2 y$, $y = C\sin(nt)$

(c) $\dfrac{d^2y}{dx^2} = \dfrac{2}{y}\left(\dfrac{dy}{dx}\right)^2$, $y = \dfrac{4}{x+1}$

Q5. [*]Cubic Solution. Find the constants a, b, c, d if $y = ax^3 + bx^2 + cx + d$ is a solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x^3 \ .$$

Q6. [*]Exponential Solution. Find the possible values of k if $f(x) = e^{kx}$ satisfies the differential equation

$$f''(x) + 3f'(x) + 2f(x) = 0 \ .$$

Solving by Direct Antidifferentiation

Q7. First Order. Find the general solution of the following differential equations using direct antidifferentiation. For those with an initial condition, find the corresponding particular solution.

(a) $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{x}}$

(b) $\dfrac{dy}{dt} = \dfrac{t^2 + 3t - 1}{t}$

(c) $\dfrac{dy}{dt} = \sin(3t + \pi)$, $y(0) = 1$

(d) $\dfrac{dy}{dx} = \dfrac{1}{2x - 1}$, $y(1) = 3$

Q8. Second Order. Find the general solution of the following by direct antidifferentiation.

(a) $\dfrac{d^2y}{dx^2} = e^{\frac{x}{2}}$

(b) $\dfrac{d^2y}{dt^2} = \sqrt{1 - t}$, $(t < 1)$

(c) $\dfrac{d^2y}{dx^2} = \dfrac{1}{(x + 1)^2}$

Separable Differential Equations

Q9. y only (autonomous DE). Solve the following separable differential equations whose right hand sides depend only on y. For those with initial conditions, find the particular solution.

(a) $\dfrac{dy}{dx} = \dfrac{1}{y^2}$

(b) $\dfrac{dy}{dx} = 1 + y^2$

(c) $\dfrac{dy}{dx} = \sqrt{y}$, $y(3) = 1$

(d) $\dfrac{dy}{dx} = y - 4$, $y(0) = 5$

Q10. Separable. Find the general solutions of the following separable differential equations. In part (e) you do not need to express x as a function of t.

(a) $\dfrac{dy}{dx} = 5y^2 \cos x$

(b) $\dfrac{dy}{dx} = e^x e^{-2y}$

(c) $\dfrac{dy}{dx} = \dfrac{\sqrt{1 - y^2}}{x}$

(d) $\dfrac{dy}{dt} = 3\sqrt{9 - y^2} \sin^4 t \cos t$

(e) $\dfrac{dx}{dt} = \dfrac{3t + e^{2t}}{x^2 + e^{-x}}$

(f) $\dfrac{dy}{dt} = \dfrac{(y^2 + 1)\cos^2 3t}{2y}$

Q11. Separable Initial Value. Find the particular solutions of the following initial value problems.

(a) $\dfrac{dy}{dx} = 3xy$, $y(0) = 3$

(b) $\dfrac{dx}{dt} = \dfrac{x}{t(t + 1)}$, $x(1) = 1$ for $t > 0$

A1. Verification.

(a) LHS $= \dfrac{dy}{dx} = \dfrac{d}{dx}[x+c] = 1 =$ RHS

(b) LHS $= \dfrac{dy}{dx} = \dfrac{d}{dx}[Ce^x] = Ce^x = y =$ RHS

(c) LHS $= \dfrac{dy}{dx} = \dfrac{d}{dx}[Ce^{2x} + 2x + 1] = 2Ce^{2x} + 2$

RHS $= 2y - 4x = 2(Ce^{2x} + 2x + 1) - 4x = 2Ce^{2x} + 2 =$ LHS

(d) LHS $= \dfrac{dy}{dx} = \dfrac{d}{dx}[Cx^{-2}] = -2Cx^{-3} = -\dfrac{2C}{x^3}$

RHS $= -2y\dfrac{1}{x} = -2(Cx^{-2})\dfrac{1}{x} = -\left(\dfrac{2C}{x^2}\right)\left(\dfrac{1}{x}\right) = -\dfrac{2C}{x^3} =$ LHS

A2. Constant Solutions.

(a) $y = 0$.

(b) No constant solutions since $e^y > 0$.

(c) No constant solutions since $y^2 + 1 > 0$.

(d) $y = \dfrac{k\pi}{2}$.

A3. Initial Value Problems. (a) $y = x + 3$ (b) $y = 2e^x$ (c) $y = e^{2x+2} + 2x + 1$ (d) $y = \frac{256}{x^2}$

A4. Second Order. Verification required.

A5. [*]Cubic Solution. $a = 1,\ b = -6,\ c = 18,\ d = -24$

A6. [*]Exponential Solution. $k = -1, -2$

A7. First Order. (a) $y = 2\sqrt{x} + C$ (b) $x = \frac{1}{2}t^2 + 3t - \log|t| + C$ (c) $y = -\frac{1}{3}\cos(3t + \pi) + \frac{2}{3}$ (d) $y = \frac{1}{2}\log(2x - 1) + 3$

A8. Second Order. (a) $y = 4e^{\frac{x}{2}} + Cx + D$ (b) $x = \frac{4}{15}(1-t)^{\frac{5}{2}} + Ct + D$ (c) $y = -\log|x+1| + Cx + D$

A9. y only (autonomous DE). (a) $y = \sqrt[3]{3(x - C)}$ (b) $y = \tan(x - C)$ (c) $y = \frac{1}{4}(x-1)^2$ (d) $y = e^x + 4$

A10. Separable. (a) $y = \frac{-1}{5\sin x + C}$ (b) $y = \frac{1}{2}\log(2e^x + D)$ (c) $y = \sin(\log|x| + C)$

(d) $y = 3\sin(\frac{3}{5}\sin^5 t + C)$ (e) $\frac{1}{3}x^3 - e^{-x} = \frac{3}{2}t^2 + \frac{1}{2}e^{2t} + C$

(f) $y = \pm\sqrt{\exp(t/2 + \sin 6t/12 + C) - 1} = \pm\sqrt{A\exp(t/2 + \sin 6t/12) - 1}$ where $A = e^C$.

A11. Separable Initial Value. (a) $y = 3e^{\frac{3}{2}x^2}$ (b) $x = \frac{2t}{t+1}$

Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.